

Mamba: Linear-Time Sequence Modeling with Selective State Spaces

May 27, 2026



State Space Model

General Sense of State Space Model



Mamba:
Linear-Time
Sequence
Modeling with
Selective
State Spaces

State Space
Model

Selective
SSMs

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State Space Models (SSMs) are sequence models that:

- Maintain a hidden state over time
- Update this state with each new input
- Generate outputs based on the state

Key idea: Compress past information into a fixed-size state.

Sentence:

"The cat that chased the mouse was hungry."

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- You don't remember every word equally
- You keep a compressed summary

SSMs do the same:

- Compress past into a fixed state
- Update it step-by-step

$$x(t) \in \mathbb{R} \quad \rightarrow \quad h(t) \in \mathbb{R}^N \quad \rightarrow \quad y(t) \in \mathbb{R}$$

- Inspired by classical dynamical systems
- Combines ideas from:
 - RNNs
 - CNNs
 - Control theory

Core Equations(Discrete Time Formulation)



$$h_t = Ah_{t-1} + Bx_t$$
$$y_t = Ch_t$$

- x_t : input
- h_t : hidden state (memory)
- y_t : output
- Equivalent to a linear RNN
- Efficient sequential computation

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Interpretation



- A : controls how memory evolves
- B : controls how input affects memory
- C : maps memory to output

Intuition:

- The model summarizes past inputs into h_t
- Updates this summary as new inputs arrive

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Continuous-Time SSM



Defined by linear differential equations:

$$h'(t) = Ah(t) + Bx(t)$$

$$y(t) = Ch(t)$$

- A : state transition
- B : input projection
- C : output projection

In this case

$$h(t) = e^{At}h(0) + \int_0^t e^{A(t-\tau)} Bx(\tau) d\tau$$

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Convolutional View



SSM can also be written as a convolution:

$$K = (CB, CAB, CA^2B, \dots)$$

$$y = x * K$$

- Equivalent to recurrence
- Enables parallel training

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Matrix expression



$$\overline{K}_j = C\overline{A}^j\overline{B}$$

$$y_t = \sum_{j=0}^t \overline{K}_j x_{t-j}$$

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_{L-1} \end{bmatrix} = \begin{bmatrix} C\overline{B} & 0 & 0 & \cdots & 0 \\ C\overline{A}\overline{B} & C\overline{B} & 0 & \cdots & 0 \\ C\overline{A}^2\overline{B} & C\overline{A}\overline{B} & C\overline{B} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ C\overline{A}^{L-1}\overline{B} & C\overline{A}^{L-2}\overline{B} & C\overline{A}^{L-3}\overline{B} & \cdots & C\overline{B} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{L-1} \end{bmatrix}$$

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Key insight: Same model, two computation paths

Convert continuous system to discrete:

$$\bar{A} = \exp(\Delta A)$$

$$\bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I)\Delta B$$

- Δ : step size
- Known as Zero-Order Hold (ZOH)

Why Discretization Matters



- Connects continuous and discrete models
- Provides:
 - Stability
 - Normalization
 - Resolution invariance
- Links to RNN gating mechanisms

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Linear Time Invariance (LTI)



- Parameters do not change over time:

(A, B, C, Δ) constant across t

- Enables:
 - Convolution representation
 - Efficient computation

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Structured SSMs



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To be efficient, impose structure on A :

- Typically diagonal

$$A \in \mathbb{R}^{N \times N}, \quad B \in \mathbb{R}^{N \times 1}, \quad C \in \mathbb{R}^{1 \times N}$$

- Reduces parameter count
- Enables fast computation

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Efficiency Challenge



Full state tensor:

$$(B, L, D, N)$$

- Very large
- Expensive in memory and compute

Core problem: How to scale SSMs efficiently

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General Definition of SSMs



SSMs broadly include:

- Kalman filters
- Hidden Markov Models
- Linear dynamical systems
- RNNs

In this paper:

- Focus on structured SSMs (S4)

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Examples built on SSMs:

- Linear Attention
- H3
- Hyena
- RetNet
- RWKV
- Different ways to integrate SSM layers
- Trade-offs in efficiency and performance

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- SSM = sequence model with latent state
- Dual view:
 - Recurrence (RNN-like)
 - Convolution (CNN-like)
- LTI enables efficiency but limits expressivity
- Structure is crucial for scalability

Next: Selective SSMs

Selective SSMs



- Motivation for Selective SSMs: 'Adapting to the Input'
- Selective SSM formulation
- Selective Scan Algorithm
- Addressing Computational Inefficiency: Hardware Scan Algorithm

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Motivation: 'Adapting to the Input'

- SSMs are **LTI** → same compression for every input.

Constant regardless of the **input**

$$\mathbf{h}_k = \bar{\mathbf{A}}\mathbf{h}_{k-1} + \bar{\mathbf{B}}\mathbf{x}_k$$

$$\mathbf{y}_k = \mathbf{C}\mathbf{h}_k$$

- However, not all inputs should be treated equally



- We need a model that can dynamically focus on important parts of input

Selective SSM (aka S6 model) should:

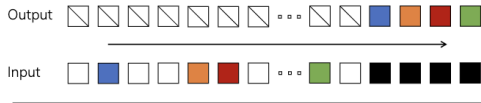
- Focus on relevant inputs and compress them into hidden state
- Ignore irrelevant inputs and filter out of hidden state

'Selection' Assessments

To assess **selectivity**, Selection can be formalized by its performance on two toy tasks:

Selective Copying - copy task that varies position of tokens to copy

Selective Copying



Example:

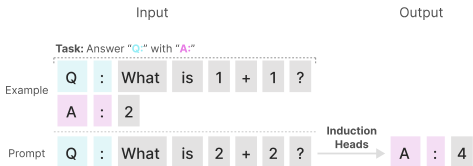


Induction heads: reproduce patterns found in input

Induction Heads



Example:



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S4: Fixed recurrence

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t$$

$$y_t = Ch_t$$

 $\bar{A}, \bar{B}, C$ fixed for all t

S6: Dynamic recurrence

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$$

$$y_t = C_t h_t$$

 Δ_t, B_t, C_t depend on x_t 7: **return** y

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Selective SSM: It Works!



By making SSM linear-time variant, selective SSM's perform well on selection tasks

MODEL	ARCH.	LAYER	Acc.
S4	No gate	S4	18.3
-	No gate	S6	97.0
H3	H3	S4	57.0
Hyena	H3	Hyena	30.1
-	H3	S6	99.7
-	Mamba	S4	56.4
-	Mamba	Hyena	28.4
Mamba	Mamba	S6	99.8

Table 1: (**Selective Copying**.)

Accuracy for combinations of architectures and inner sequence layers.

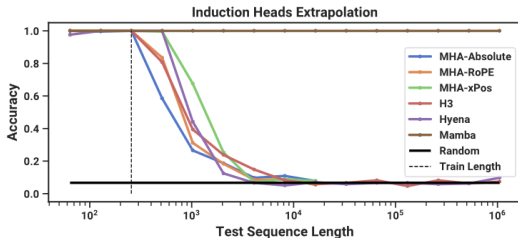


Table 2: (**Induction Heads**.) Models are trained on sequence length $2^8 = 256$, and tested on increasing sequence lengths of $2^6 = 64$ up to $2^{20} = 1048576$. Full numbers in Table 11.

Figure: Gu and Dao, 2024

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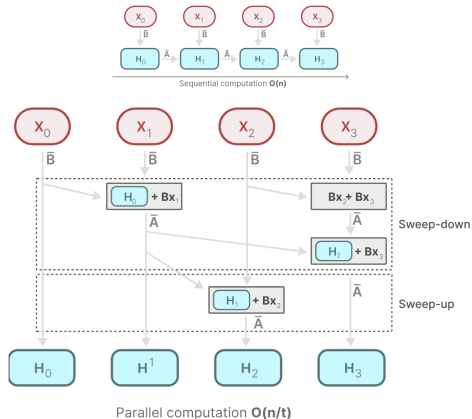
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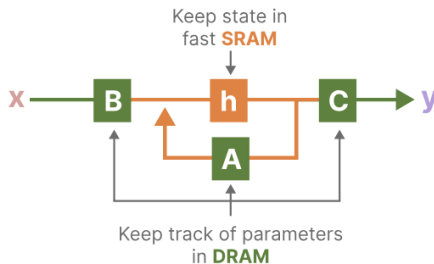
Mamba's Hardware-Aware Algorithm - An Efficient Implementation for Selective SSM

Mamba implements kernel fusion, parallel scan, and recomputation to address the computational inefficiencies of selective scan:

'Parallel' Scan



Kernel fusion



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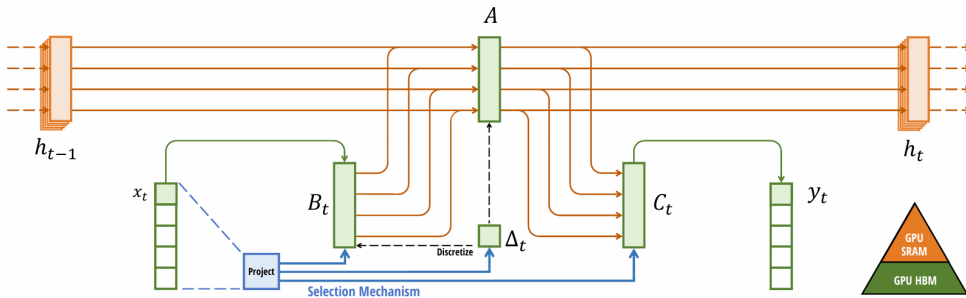
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Result: Select SSM with Hardware-Aware State Expansion



Selective State Space Model *with Hardware-aware State Expansion*



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Mamba Architecture



- Combine:
 - SSM block
 - MLP block
- Single unified block
- Repeated across layers
- Use SiLU activation
- Residual + normalization
- Similar parameter count to Transformer

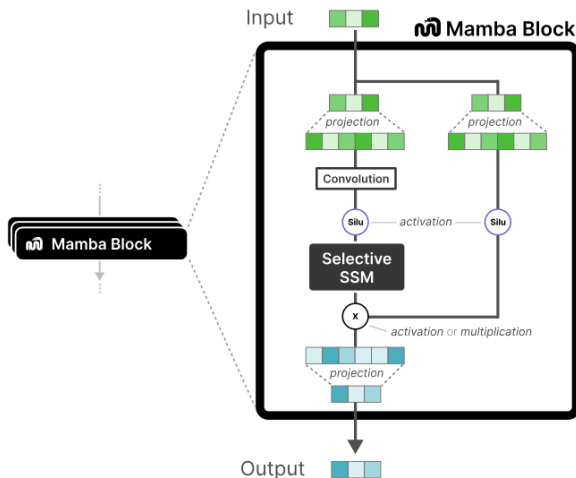
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Architecture Details



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Diagram of Selective SSM

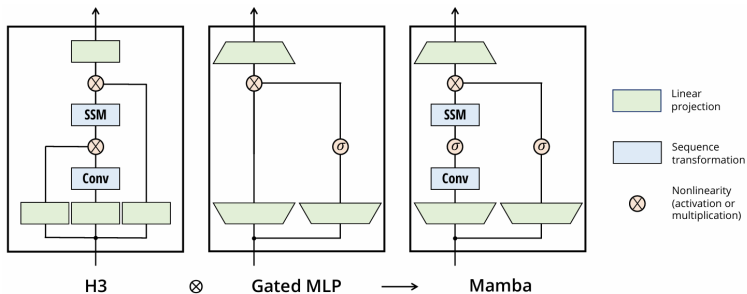


Figure: Mamba Structure

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Connection to Gating



Special case:

$$g_t = \sigma(\text{Linear}(x_t))$$

$$h_t = (1 - g_t)h_{t-1} + g_tx_t$$

- Equivalent to gated RNN

Insight: selection generalizes gating

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Interpretation of Δ



- Controls memory vs input
- Large Δ :
 - Reset state
 - Focus on input
- Small Δ :
 - Keep memory
 - Ignore input

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Step 3: Elementwise gating

$$h = g \odot v \quad h_{b,t,i} = g_{b,t,i} \cdot v_{b,t,i}$$
$$y = W_o h$$
$$y = W_o(\sigma(W_u x) \odot W_v x)$$

Input: $x \in \mathbb{R}^{B \times L \times D}$

Step 1: Linear projections

$$u = W_u x, \quad v = W_v x$$

Step 2: First gating

$$h_1 = u \odot v$$

Step 3: Local convolution

$$h_2 = \text{Conv}(h_1) = \sum_{k=0}^{K-1} w_k^{(i)} \cdot h_{1,b,t-k,i} \quad K \text{ as the kernel size}$$

Step 4: SSM transformation

$$h_3 = \text{SSM}(h_2) \quad (s_{b,t,i} = \bar{A}_i s_{b,t-1,i} + \bar{B}_i h_{2,b,t,i}, \quad h_3 = C_i s_{b,t,i})$$

Step 5: Second gating

$$h_4 = h_3 \odot v$$

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Step 6: Output projection

$$y = W_o h_4$$

Final form:

$$y = W_o \left(\text{SSM}(\text{Conv}(W_u x \odot W_v x)) \odot W_v x \right)$$

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Step 3: Gated interaction

Step 4: Local convolution

Step 5: Selective SSM

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Step 6: Output activation

$$h_4 = \sigma(h_3)$$

Step 7: Output projection

$$y = W_o h_4$$

Final form:

$$y = W_o \left(\sigma \left(\text{SSM}_{\text{sel}} \left(\text{Conv} \left(\sigma(W_u x) \odot W_v x \right) \right) \right) \right)$$

Interpretation of B and C



- B_t : controls input $\rightarrow$ state
- C_t : controls state $\rightarrow$ output
- Enables fine-grained filtering

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Why Selection Works



- Filters irrelevant tokens
- Adapts to content
- Enables long-context reasoning

Key benefit:

- Efficient + expressive

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